

Assessing prediction errors of generalized tree biomass and volume equations for the boreal forest region of west-central Canada

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Abstract: Aboveground tree biomass and volume are required inputs to models that estimate carbon budgets and ecosystem productivity. Generalized equations are often used to estimate biomass and volume when local equations are unavailable. This study determined whether there was a concomitant increase in prediction error from increasing levels of equation generalization. Local site, generalized regional, and generalized national allometric equations were compared for 10 species distributed across 119 sites in a region defined by west-central Canada. This study employed regression fit statistics and two prediction error metrics, the average prediction error (APE) from the prediction error sum of squares (PRESS) statistic and mean prediction bias. The APE was 9, 12, and 25 kg of biomass per tree for local site, generalized regional, and national equations, respectively. The mean prediction bias for biomass and volume were statistically similar between local level and generalized regional equations across all species. Predictions from generalized national equations were statistically similar for 5 of 10 species when compared with those from local site and generalized regional equations. While local site equations were most accurate for a given site, results indicate that generalized regional equations will produce reasonable estimates of biomass and volume at sites in this region of Canada.

Résumé : Le volume et la biomasse aérienne des arbres sont des intrants essentiels dans les modèles d'estimation du bilan du carbone et de la productivité des écosystèmes. Des équations généralisées sont souvent utilisées pour estimer la biomasse et le volume lorsque les équations locales ne sont pas disponibles. Cette étude détermine s'il y a une augmentation de l'erreur de prédiction causée par l'augmentation concomitante du degré de généralisation des équations. Les équations allométriques nationales généralisées, les équations régionales généralisées et les équations locales de la station ont été comparées pour 10 essences dans 119 stations distribuées à travers le centre-ouest du Canada. Cette étude utilise les statistiques d'ajustement par régression et deux mesures des erreurs de prédiction, l'erreur moyenne de prédiction (EMP) à partir de la statistique PRESS et le biais moyen de prédiction. L'EMP atteint respectivement 9, 12 et 25 kg de biomasse par arbre pour les équations locales, régionales généralisées et nationales généralisées. Le biais moyen de prédiction des équations locales et régionales généralisées pour la biomasse et le volume est similaire pour toutes les essences. Les prédictions obtenues avec les équations nationales généralisées sont statistiquement similaires à celles des équations locales et régionales généralisées pour 5 des 10 essences. Alors que les équations locales sont les plus précises pour une station donnée, les résultats indiquent que les équations régionales généralisées produiront des estimations raisonnables de la biomasse et du volume pour des stations dans cette région du Canada.

[Traduit par la Rédaction]

Introduction

Aboveground forest biomass, the dry mass of live plant material in trees and understory species, is an indicator of how much carbon (C) exists in a forest ecosystem. As a result, information about forest biomass has been used in studies of sustainable forest management (Canadian Council of Forest Ministers 2003), projections of ecosystem productivity (Bernier et al. 1999), quantifying energy and nutrient flows (Schroeder et al. 1997), and providing inputs to models that calculate and forecast C budgets (Kurz and Apps

1999), amongst others. Accurate estimates of C stocks and their changes are necessary to determine the potential impacts of global change on forests, and more locally, to assess changes in forest structure that may result from disturbance, successional processes, and management practices (Schroeder et al. 1997).

Estimating aboveground forest biomass starts with estimates of biomass at the tree level that are summed across individual trees on a plot and, subsequently, scaled to a per unit area basis at the stand level (Fournier et al. 2003; Jenkins et al. 2003). It is these stand level estimates that provide the basis for its scaling to a remote sensing image from which mapping across the landscape can be achieved (Hall et al. 2006; Luther et al. 2006). Consequently, accurate estimation of tree biomass becomes fundamentally important to scaling and mapping at stand and landscape levels (Fournier et al. 2003). There is a strong and consistent allometric relationship between diameter at breast height (DBH) and aboveground tree biomass that forms the basis for most published biomass estimation equations, although they may also

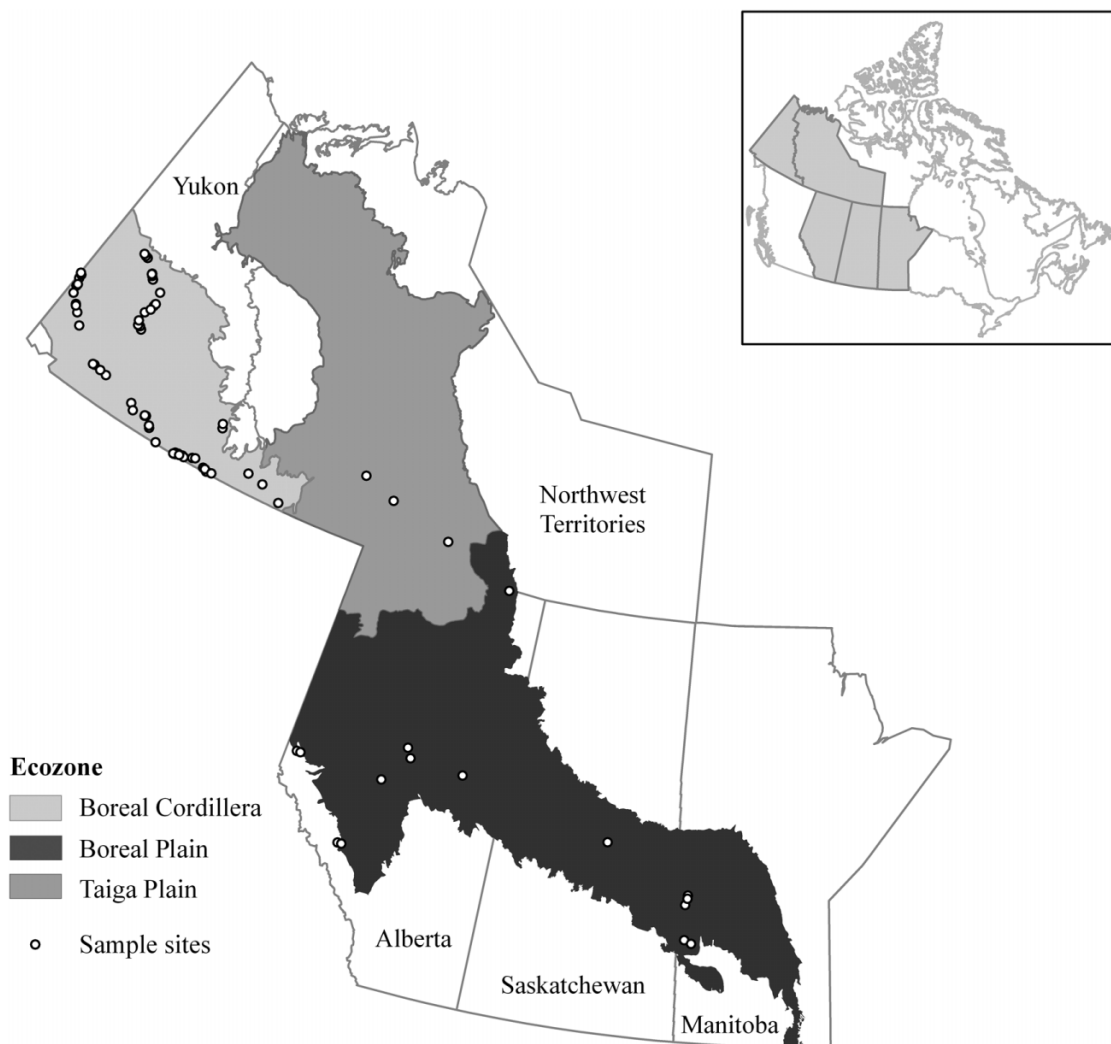
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Fig. 1. The distribution of biomass sample plots across west-central Canada by boreal forest ecozone. Inset map shows the location of the study region in Canada.

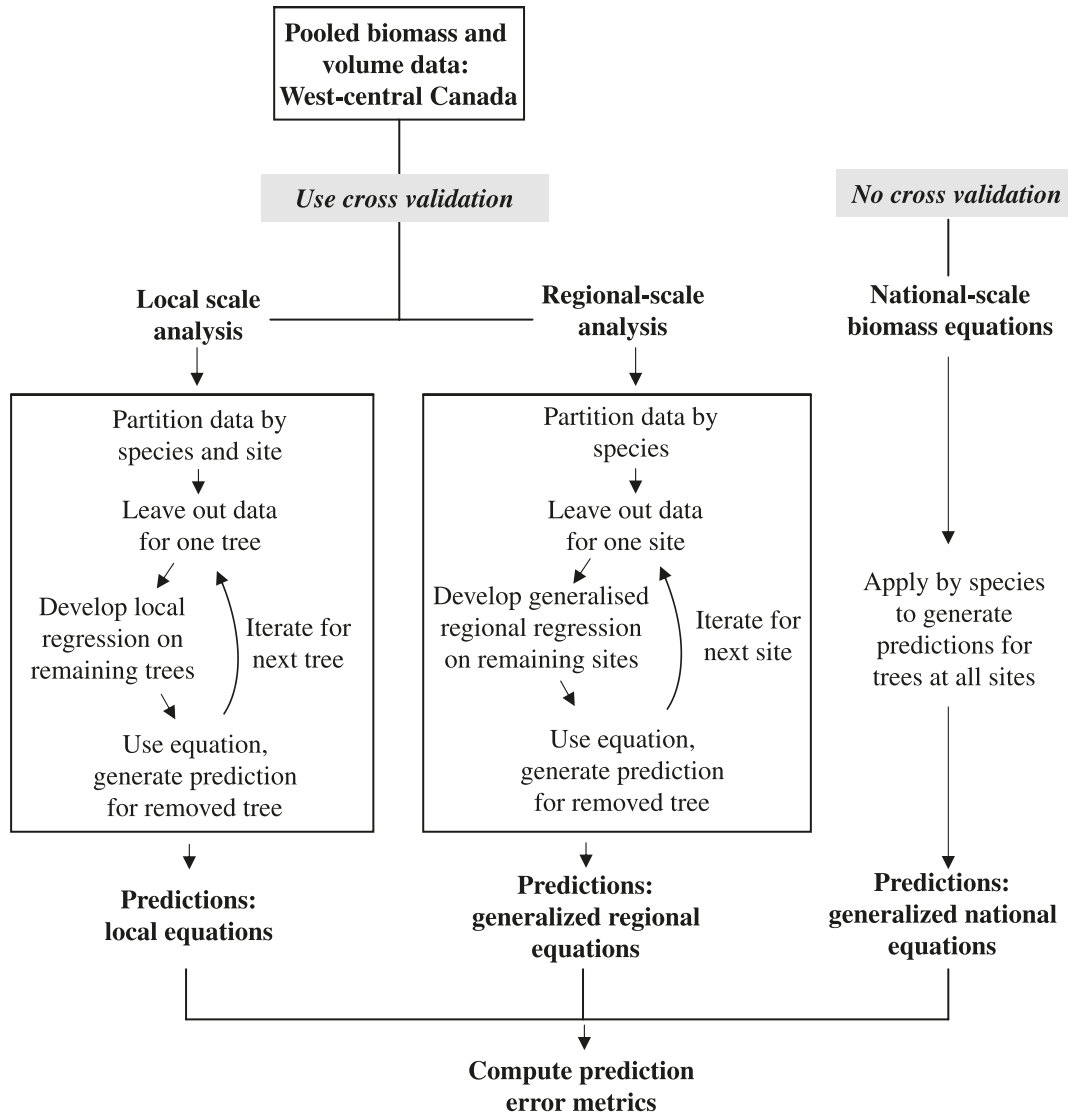


include measures of tree height and (or) tree form (Brown 2002; Chave et al. 2005). Similarly, allometric relationships are used to estimate tree volume, which is inherently correlated with tree biomass (Smith et al. 2003). Volume inventory data are widely collected and available in many parts of the world, more so than biomass, and for this reason it has been used as a data source for estimating biomass (Schroeder et al. 1997). Stand-specific volume equations are not suited for regional estimates, however, resulting in a need for generalized volume equations (Muukkonen 2007). Thus, there is value in deriving and evaluating generalized equations that estimate both volume and biomass.

There is a plethora of tree biomass and volume estimation equations in the published literature (Ter-Mikaelian and Korzukhin 1997; Lambert et al. 2005). Some published equations have been created specifically for localized sites that are then most suited to tree populations at those sites or at sites of similar ecological, environmental, and stand-structure conditions (Wang et al. 2002). Because the collection of tree measurements for the development of biomass and volume equations can be a time consuming and costly

exercise, readily available, published equations are frequently used to generate estimates at new locations (Fazakas et al. 1999; Fournier et al. 2003). Although equations are often reported with relatively strong relationships and high coefficient of determination (R^2) values (Ter-Mikaelian and Korzukhin 1997), there is no guarantee of accurate estimates when a given equation is applied to locations outside of the geographic region sampled for equation development. To broaden the geographic region over which equations can be applied, generalized equations, particularly for biomass, have been developed for different species in North America (Schroeder et al. 1997; Lambert et al. 2005). Generalized biomass equations are intended to provide realistic and consistent biomass estimates (Jenkins et al. 2003) over a specific range of environmental and site factors and tree sizes, such that estimation errors should fall within the range of variation in the data used to develop the generalized equations (Brown 2002).

To what extent could generalized equations accurately estimate biomass and volume given the inherent variability in environmental and ecological conditions across large geo-

Fig. 2. Conceptual outline of the methods used for prediction error analyses.

graphic regions? Local equations developed at a specific site are presumed to generate more accurate estimates at that site than those from generalized regional equations, and estimates from generalized regional equations should be more accurate than those from generalized national equations. Of interest was determining if estimates among local, generalized regional, and generalized national equations differed in prediction errors across different species and sites, within the applicable geographic extent for which the equations were originally developed.

Few studies have attempted to quantify the level of prediction error incurred with the use of generalized biomass equations at a site or tree scale, particularly in comparison with local equations. Understanding the magnitude of differences in predictions would more clearly identify the benefits and issues associated with the application of generalized equations for estimating tree biomass and volume. Knowledge of prediction errors aids in formulating error budgets and in assessing the propagation of errors from the tree to stand, region, or national levels (Phillips et al. 2000). Goodness-of-fit statistics, such as R^2 and root mean square error (RMSE),

are often used to report regression model performance. While these statistics are an indicator of model performance, additional measures of model performance are often employed (Yang et al. 2004). Robust calculations of prediction error variability can be determined with resampling methods such as cross-validation of raw data (Efron and Tibshirani 1993) or by validating existing equations against independent data. Such data are seldom available because of the cost and logistics associated with collecting data sets of sufficient sample size that would be representative over large geographic regions.

To address the issue of prediction errors, this study offers an update to the publication of generalized biomass equations for the boreal forest region of west-central Canada (Singh 1986), expands the application to tree volume, estimates prediction errors using a cross-validation statistic based on the prediction error sum of squares (PRESS), and compares these regional equations with local equations and generalized national equations recently published by Lambert et al. (2005). The specific objectives of this study were (1) to create local and generalized regional equations

Table 1. Descriptive statistics for the distribution of the number of sample trees per site and associated tree dbh, aboveground tree biomass, and aboveground total tree volume by species used in modeling biomass and volume.

Species	Total no. of sample trees	No. of sample trees (per site)			DBH (cm)			Biomass (kg)			Volume (m ³)		
		Mean ± SD	Range		Mean ± SD	Range		Mean ± SD	Range		Mean ± SD	Range	
White spruce	415	14±7	7–30		18.1±10.4	2.0–45.3		156.6±198.9	0.53–666.1		0.33±0.47	0.0008–3.89	
Black spruce	383	15±7	6–36		14.1±7.1	3.4–38.9		79.8±99.6	1.6–631.4		0.14±0.18	0.0008–1.03	
Lodgepole pine	172	11±7	6–30		15.4±8.4	2.5–37.2		108.0±129.3	0.92–582.1		0.21±0.27	0.0011–1.32	
Jack pine	112	16±7	9–29		20.0±10.0	2.4–54.3		174.8±165.2	0.89–1378.9		0.33±0.32	0.0026–1.33	
Tamarack	112	19±4	12–23		18.0±9.1	2.5–39.0		127.7±124.7	0.83–720.7		0.21±0.20	0.0011–1.03	
Alpine fir	60	30±14	20–40		19.6±10.2	1.8–39.0		167.1±160.9	0.67–638.3		0.30±0.30	0.0005–0.95	
Balsam fir	60	15±6	8–20		20.1±10.9	3.0–38.0		186.7±182.9	1.5–505.4		0.36±0.36	0.0005–1.43	
Trembling aspen	266	15±7	6–33		16.0±9.2	2.1–44.3		124.7±162.9	0.29–822.9		0.24±0.32	0.0012–1.72	
Balsam poplar	101	15±8	7–31		19.4±10.5	1.5–39.0		141.1±143.0	0.36–897.5		0.32±0.31	0.0004–1.24	
White birch	55	11±3	8–15		18.4±11.2	2.1–36.5		225.1±258.9	1.7–491.0		0.25±0.24	0.0004–1.05	

that estimate total tree biomass and volume for 10 tree species that occur within the Boreal region of west-central Canada; (2) to determine if there are differences in predictions of tree biomass and volume from (i) local, (ii) regional, and (iii) published national equations.

Methods

Data

Individual tree biomass and volume data were collected from the Boreal Cordillera, Boreal Plain, Boreal Shield, and Taiga Plains Ecozones (Ecological Stratification Working Group 1995) as part of the Energy from the Forest research project (ENFOR) of the Canadian Forest Service that was undertaken across Canada during the 1980s (Lambert et al. 2005). For this study, we used a subset of this national data set that covers the Prairie provinces (Singh 1982), Northwest Territories (Singh 1984), and Yukon Territory (Manning et al. 1984). Approximately 51% of Canada's boreal forest is located within these provinces and territories within which 70% was represented by the three ecozones that were sampled for this study (Fig. 1). Within this study region, there is a considerable range in terrain and landform comprising level to rolling terrain in the Boreal Plain to hummocky terrain in the Boreal Shield and mountain ranges with incidences of permafrost in the Boreal Cordillera. This variability results in a considerable diversity in vegetation composition and structure, with the 10 most predominant tree species consisting of white spruce (*Picea glauca* (Moench) Voss), black spruce (*Picea mariana* (Mill.) BSP), lodgepole pine (*Pinus contorta* Dougl. ex. Loud. var. *latifolia* Engelm.), jack pine (*Pinus banksiana* Lamb.), tamarack (*Larix laricina* (Du Roi) K. Koch), alpine fir (*Abies lasiocarpa* (Hook.) Nutt.), balsam fir (*Abies balsamea* (L.) Mill.), trembling aspen (*Populus tremuloides* Michx.), balsam poplar (*Populus balsamifera* L.), and white birch (*Betula papyrifera* Marsh.) in varying amounts.

Total aboveground biomass and total tree volume data were collated for each tree species. The data contained dry mass (kg) biomass measurements for a range of tree components including the stem, branch, foliage, and bark, in addition to volume (m³) measurements for the merchantable stem and nonmerchantable stem (not including the stump) and top components of the tree. Component data were summed to compute total biomass (kg·tree⁻¹) and volume (m³·tree⁻¹) for each individual tree. The data set consisted of 1955 sample trees that represented the 10 species at 228 unique locations (i.e., latitude and longitude coordinates) across the study area, resulting in biomass and volume data for 147 unique species-by-location combinations (hereafter called "sites"). Of these sites, only those that contained five or more trees by species were selected for modeling, which resulted in a data set consisting of 1736 trees at 119 sites (hereafter called the "full data set").

Creation of local and generalized regional equations for west-central Canada

To meet the first objective and based on the full data set, local and generalized regional biomass and volume equations were created for each of the 10 species based on the log-transformed power model (Chave et al. 2005):

Table 2. Generalized regional regression equations and statistics (fit index (FI), standard error of estimate (SEE), and bias correction factor (CF)) for the west-central Canada region for aboveground tree biomass (kg) and tree volume (m³) by species, based on the equation form: $\ln Y = b_0 + b_1(\ln \text{DBH})$, generated from the full data set of all trees by species.

Species	Biomass (dependent variable Y, kg)					Volume (dependent variable Y, m ³)				
	<i>b</i> ₀	<i>b</i> ₁	FI ^a	SEE ^a	CF ^b	<i>b</i> ₀	<i>b</i> ₁	FI ^a	SEE ^a	CF ^b
White spruce	−2.464	2.366	0.94	35.9	1.033	−9.267	2.623	0.95	0.098	1.02
Black spruce	−2.276	2.371	0.93	43.5	1.024	−9.267	2.597	0.96	0.036	1.016
Lodgepole pine	−2.021	2.274	0.94	25.2	1.019	−9.183	2.611	0.95	0.054	1.022
Jack pine	−2.236	2.355	0.95	42.2	1.015	−8.762	2.441	0.90	0.102	1.028
Tamarack	−2.463	2.446	0.97	22.6	1.019	−9.167	2.509	0.95	0.046	1.018
Alpine fir	−2.007	2.279	0.97	30.4	1.018	−9.544	2.664	0.97	0.050	1.01
Balsam fir	−2.341	2.372	0.97	20.0	1.020	−9.206	2.581	0.94	0.086	1.027
Trembling aspen	−2.763	2.524	0.94	39.7	1.022	−9.037	2.544	0.95	0.068	1.017
Balsam poplar	−2.138	2.430	0.96	51.7	1.013	−9.229	2.566	0.93	0.082	1.019
White birch	−2.184	2.325	0.97	29.9	1.018	−8.767	2.415	0.92	0.121	1.031

^aFor regressions using a log-transformed independent variable: FI and SEE are analogous to the *R*² and root mean square error (RMSE) computed from parametric regression, respectively (Parresol 1999).

^bCorrection for bias resulting from the log transformation of the independent variable (DBH), calculated as one-half the transformed regression error variance (Baskerville 1972).

[1] $\ln(Y) = a + b \ln(\text{DBH})$

where *Y* is either total aboveground tree biomass (kg·tree^{−1}) or total tree volume (m³·tree^{−1}). Local equations were created using data at each site location, resulting in 119 separate biomass and volume equations. A generalized, regional biomass and volume equation was created for each of the 10 species from data pooled across all sites. For each of the local and regional equations, a regression fit index (FI) and standard error of estimate (SEE) was computed using data back-transformed from logarithmic to real units based on equations reported by Parresol (1999). The FI and SEE are statistical measures analogous to the coefficient of determination (*R*²) and RMSE, respectively, that are often reported from linear regression with nontransformed data (Parresol 1999). To correct for bias introduced from regression equations based on the log transformation of the independent variable, a bias correction factor (CF) was computed for each equation as one-half the transformed regression error variance (Baskerville 1972). The FI and SEE statistics were used to compare overall model fit between local and regional equations, and these values were, in turn, compared with the model fit statistics reported for the generalized national equations (Lambert et al. 2005).

Analysis of prediction differences among local, regional, and national equations

Our second objective was to assess differences in model predictions of biomass and volume when estimates from local sites were compared with generalized regional and national equations. Biomass and volume predictions were generated for both the local and regional equations using a cross-validation process on the full data set for west-central Canada. The employment of cross-validation results in data that can be used to generate more robust and unbiased estimates of prediction error when independent data of sufficient size to evaluate model performance are not available (Efron and Tibshirani 1993). Local and regional equation predictions were then compared with those generated from direct application of published national biomass equations

(Lambert et al. 2005) to our west-central Canadian data set for each species.

Generating biomass and volume predictions

Local and generalized regional equations:

At the local scale, regression equations of the form described in eq. 1 were fit using cross-validation methods that involved iteratively removing each sample tree at a given site, fitting regression eq. 1 to the remaining trees at that site, and generating a cross-validated prediction for biomass or volume for the removed tree (Fig. 2a). The cross-validation analysis resulted in a set of predictions for both the biomass and volume of all trees at each of the 119 site locations. Log-transformed predictions were ultimately back-transformed to original units after correction for bias introduced from the logarithmic model.

Generalized regional biomass and volume predictions were similarly derived using a cross-validation procedure (Fig. 2b). Biomass and volume data from the full data set were initially subset by each of the 10 species. For each species, data from one site location (where each site consists of five or more trees) were removed. The data for the remaining sites were pooled and used to construct a generalized equation of the form presented as eq. 1. The resultant generalized equation was used to generate a set of predictions for trees that occurred in the removed site location. This procedure was repeated for each unique site within the species subset and, subsequently, for each of the 10 species, resulting in a set of cross-validated predictions for each of the 119 sites.

Generalized national equations:

The national biomass equations for Canada (Lambert et al. 2005) were used to generate biomass predictions for each tree at the 119 sites by direct application of the equations (Fig. 2c). The national equation set contains, for each species, independent, additive equations for each of bark, stem, branch, and foliage components, of which its sum was used to estimate total aboveground tree biomass (Lambert et

Table 3. Across-site means (± 1 SE), by species, of average prediction error (APE) and mean prediction bias statistics generated from the assessment of prediction errors (Fig. 2) using site and generalized biomass and volume equations.

Species	No. of sites	Biomass (kg)			Volume (m ³)		
		Local, site			Generalized, national		
		APE (mean $\pm$ SE)	Bias (mean $\pm$ SE)	APE (mean $\pm$ SE)	APE (mean $\pm$ SE)	Bias (mean $\pm$ SE)	APE (mean $\pm$ SE)
White spruce	29	9.2 $\pm$ 9.4	0.5 $\pm$ 2.3	14.4 $\pm$ 17.9	27.8 $\pm$ 27.1	5.7 $\pm$ 20.9	0.036 $\pm$ 0.045
Black spruce	26	5.6 $\pm$ 8.7	0.7 $\pm$ 1.9	9.1 $\pm$ 12.0	15.1 $\pm$ 11.9	-1.5 $\pm$ 10.9	0.019 $\pm$ 0.037
Lodgepole pine	15	10.5 $\pm$ 15.0	-0.4 $\pm$ 2.7	7.0 $\pm$ 10.5	18.1 $\pm$ 18.4	4.8 $\pm$ 13.1	0.020 $\pm$ 0.032
Jack pine	7	6.3 $\pm$ 5.4	0.8 $\pm$ 1.4	25.0 $\pm$ 26.4	34.7 $\pm$ 22.7	-2.7 $\pm$ 25.9	0.018 $\pm$ 0.019
Tamarack	6	5.9 $\pm$ 6.4	-0.4 $\pm$ 1.5	5.6 $\pm$ 2.0	27.8 $\pm$ 7.7	-13.6 $\pm$ 3.5	0.024 $\pm$ 0.021
Alpine fir ^a	2	29.0	5.5	25.2	28.5	4.6	0.067
Balsam fir	4	23.2 $\pm$ 16.9	4.8 $\pm$ 3.8	11.3 $\pm$ 13.9	40.7 $\pm$ 5.9	25.0 $\pm$ 17.5	0.053 $\pm$ 0.043
Trembling aspen	18	8.8 $\pm$ 11.8	-0.1 $\pm$ 2.8	13.0 $\pm$ 13.3	27.5 $\pm$ 15.1	-10.0 $\pm$ 12.3	0.024 $\pm$ 0.029
Balsam poplar	7	16.7 $\pm$ 26.8	2.7 $\pm$ 4.6	7.4 $\pm$ 8.5	38.6 $\pm$ 26.0	-18.0 $\pm$ 13.5	0.018 $\pm$ 0.021
White birch	5	7.7 $\pm$ 4.9	2.3 $\pm$ 1.4	7.6 $\pm$ 6.5	33.8 $\pm$ 22.5	12.7 $\pm$ 11.7	0.005 $\pm$ 0.006
Overall mean	119	9.3 $\pm$ 12.4	0.8 $\pm$ 2.7	11.6 $\pm$ 14.7	25.5 $\pm$ 20.7	-0.3 $\pm$ 17.7	0.025 $\pm$ 0.035

Note: National-scale volume equations were not available, limiting the volume analysis to a comparison between site and generalized regional equations.

^aInsufficient number of sites available to calculate SE values for this species.

al. 2005). Model parameters for volume were not available at the national level resulting in the analysis of differences in predictions being confined to tree biomass only at this scale of equation generalization.

Assessing biomass and volume equations

A variety of prediction error metrics and tests have been reported in the published literature to assess the prediction performance of allometric models (Parresol 1999; Yang et al. 2004). Employing various statistical measures of model performance and validation are generally described as attempts to ascertain whether or not a model is an acceptable representation of reality (Yang et al. 2004). In this study, in addition to the general assessment of model fit statistics, two error metrics were employed to quantify the overall differences among predictions derived from local, generalized regional, and generalized national equations at each of the 119 sites: (i) PRESS from which an estimate of average prediction error (APE) (Weisberg 1985) was computed; and (ii) mean bias computed as an overall average and summarized by diameter class similar to that employed by Zhang (1997). All statistical computations in this study were undertaken with the R statistical software version 2.4.0 (R Development Core Team 2006).

Average prediction error (APE):

At each site, APE was computed from the PRESS statistic as (Weisberg 1985, p. 230)

$$[2] \quad APE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}}$$

where y_i is the actual biomass or volume for sample i , $\hat{y}_i$ is the biomass or volume prediction for sample i , and n is the number of trees sampled at a given site.

Mean bias:

The mean prediction bias ($\overline{\text{Bias}}$) was calculated as the mean difference between actual values and predicted values for each tree selected in the leave-one-out cross-validation exercise (Zhang 1997).

$$[3] \quad \overline{\text{Bias}} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)}{n}$$

The most accurate equations should produce $\overline{\text{Bias}}$ values that are near zero; positive $\overline{\text{Bias}}$ values constitute an under-prediction, whereas negative $\overline{\text{Bias}}$ values constitute an over-prediction. We also used a one-way analysis of covariance (ANCOVA) with tree DBH as the covariate to test for differences in $\overline{\text{Bias}}$ from application of local, regional, and national equations to each of the 10 species. Where the "equation" term was significant, Bonferroni multiple comparisons were employed to determine which equation pairs produced $\overline{\text{Bias}}$ values that were statistically different. For tree volume, comparisons could only be made between local and regional generalized equations because national generalized volume equations were not available. All statistical tests in this study were conducted at the 5% level of significance.

Table 4. Results from analysis of covariance to test for statistical differences in mean prediction bias, by species, for biomass predictions generated from the use of (1) local site, (2) generalized regional, and (3) generalized national equations at sites across western Canada.

Species	<i>p</i> value	Bonferroni comparisons ^a
White spruce	0.23	No significant differences
Black spruce	0.000*	2–3, <i>p</i> < 0.001*
Lodgepole pine	0.22	No significant differences
Jack pine	0.05	No significant differences
Tamarack	0.000*	1–3, <i>p</i> < 0.001*; 2–3, <i>p</i> < 0.001*
Alpine fir	0.41	No significant differences
Balsam fir	0.004*	1–3, <i>p</i> = 0.01*; 2–3, <i>p</i> = 0.02*
Trembling aspen	0.000*	1–3, <i>p</i> = 0.01*; 2–3, <i>p</i> < 0.001*
Balsam poplar	0.000*	1–3, <i>p</i> < 0.001*; 2–3, <i>p</i> < 0.001*
White birch	0.24	No significant differences

Note: Bonferroni multiple comparisons were employed when significant differences were generated from the analysis of covariance to identify the equations for which prediction bias was statistically different. *, statistically significant at *p* = 0.05.

^a1, local site; 2, regional generalized; and 3, national generalized.

The magnitude of $\overline{\text{Bias}}$ is notably influenced by the size of the tree. To provide further insights into the relative distribution of $\overline{\text{Bias}}$ by species, the *n* in eq. 3 was replaced by the number of trees per 5 cm diameter class so that the distribution of prediction bias by tree size could be derived for each species.

Results

There was a relatively wide range in tree sizes sampled across all species with tree biomass being lowest for black spruce and highest for white birch (Table 1). The number of species sampled was somewhat influenced by its geographic range. White spruce, black spruce, and trembling aspen, for example, had the largest number of trees sampled, but they are also tree species that had the widest geographic range of the 10 species within the study area (Ecological Stratification Working Group 1995). While a more equitable number of trees sampled for each species would have been preferable, the numbers compiled from existing tree biomass data sets did provide a range of biomass and volume from which to compare predictions from local, regional, and national equations.

The FI values across all species for generalized regional biomass and volume equations were consistently high across all species (FI > 0.90, Table 2). The SEE values ranged from 20 to 52 kg·tree⁻¹ for biomass and 0.036 to 0.121 m³·tree⁻¹ for volume (Table 2). These results were within the range of the 119 individual local biomass and volume equations generated as part of this study (FI biomass, 0.51–0.99; FI volume, 0.71–0.99; SEE biomass, 1–109 kg·tree⁻¹; and SEE volume, 0.001–0.147 m³·tree⁻¹) (unpublished data), and were comparable to *R*² and RMSE results presented by Lambert et al. (2005) for the generalized national biomass equations. While there was an apparent strong statistical association between DBH and biomass, further assessment of prediction errors was needed to provide insights as to how well generalized equations could estimate biomass or volume at new sites, relative to local equations.

The APE for biomass, across all sites and species, generally increased in the order of local equations (9.3 kg·tree⁻¹) being less than the regional equations (11.6 kg·tree⁻¹), and both local and regional equations being considerably less than the national equations (25.5 kg·tree⁻¹) (Table 3). For 5 of the 10 species examined (Table 3, lodgepole pine, tamarack, alpine fir, balsam fir, and balsam poplar), APE values for regional equations were lower than those for local site equations. This result may be attributed to the relatively low number of sample trees at some sites (Table 1) such that leaving out one tree with each cross-validation iteration at the local analysis scale can have a significant influence on DBH–biomass regression relationships, thereby possibly inflating the APE values (Table 3). Inferences could also be derived from assessing the variability about the APE values. Standard errors were smaller for local site equations than for the generalized equations and, for some species (eg., white spruce, black spruce, lodgepole pine, jack pine, and trembling aspen), estimates between generalized regional and national equations were not largely different when their standard errors were considered (Table 3). Based on these results, biomass prediction errors increased with increasing levels of equation generalization, and volume estimates between local site and generalized regional equations were similar, with exceptions that may be attributed to the sample size distribution of trees and sites among the 10 species.

From local site to generalized regional equations, $\overline{\text{Bias}}$ increased, and then further increased for 7 of 10 species from generalized regional to national equations (Table 3). The exceptions were alpine fir $\overline{\text{Bias}}$, whose value was similar between regional and national equations, and black spruce and jack pine $\overline{\text{Bias}}$, whose values from the generalized national equations were smaller than those from the regional biomass estimates. Local equations produced largely nonbiased biomass predictions and were less variable than those exhibited by the regional and national equations across most species. The regional equations underestimated biomass for both jack pine and balsam fir, whereas national equations underestimated balsam fir, white birch, white spruce, and lodgepole pine and overestimated balsam poplar, tamarack, and trembling aspen (Table 3). The direct application of the generalized national equations from Lambert et al. (2005) to the site data resulted in prediction bias values that were quite variable among species, with both over- and under-estimations that tended to average out near zero. For volume, there was a very slight overestimation from both local site and regional equations as exhibited by the negative $\overline{\text{Bias}}$ values (Table 3). Otherwise, $\overline{\text{Bias}}$ from volume predictions were comparable between local and generalized regional equations (Table 3).

There were no statistical differences in $\overline{\text{Bias}}$ among local, regional, and national biomass equations for white spruce, lodgepole pine, jack pine, alpine fir, and white birch (Table 4). However, there were statistical differences in $\overline{\text{Bias}}$ among equations for the remaining five species: trembling aspen, balsam fir, tamarack, balsam poplar, and black spruce (Table 4). For these species, multiple comparison test results reported $\overline{\text{Bias}}$ was statistically similar between local and regional equations, whereas it was statistically different from the national equation $\overline{\text{Bias}}$ values (Table 4). There were no statistical differences in volume $\overline{\text{Bias}}$ between the

Fig. 3. Mean prediction bias plotted against 5 cm DBH classes across sites in the study region, by species, for local, site (-■-), generalized regional (-△-), and generalized national (-○-) equations, for (a) biomass and (b) volume. For trees in each DBH class, bias was calculated as the mean difference between actual and predicted biomass and volume. National volume equations were not available, thus, limiting the comparison between local and generalized regional equations.

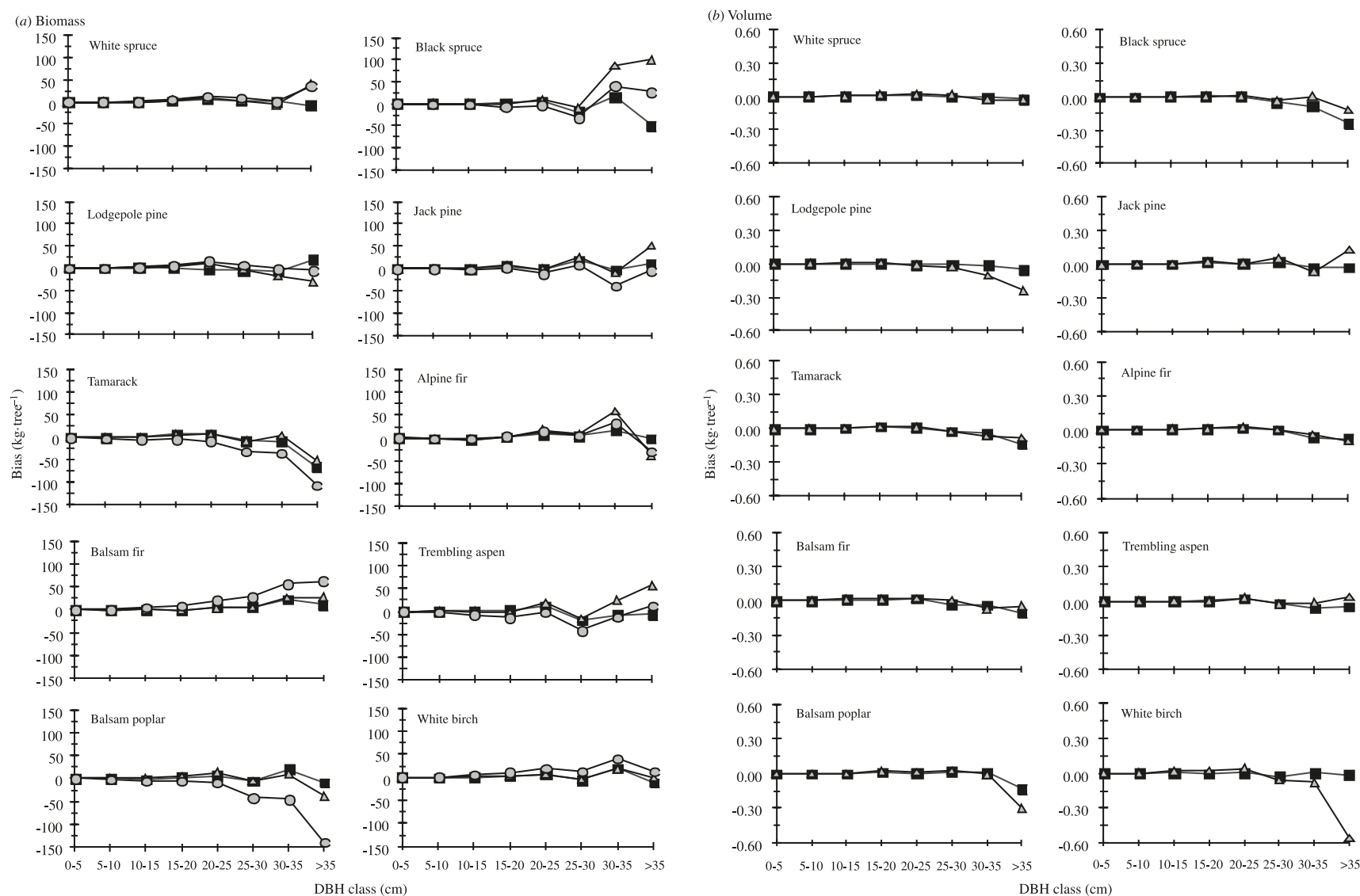


Table 5. Potential biological, environmental, and data-modeling factors affecting tree allometry and related estimation equations of above-ground tree biomass and total volume as collected from the literature.

Factors	Literature source
Biological factors	
Tree species or species group	Ter Mikaelian and Korzukhin (1997); Lambert et al. (2005)
Tree size range or class	Schroeder et al. (1997); Jenkins et al. (2003)
Tree structure and height	Vallet et al. (2006)
Wood density	Chave et al. (2005)
Stand composition and structure	Bond-Lamberty et al. (2002)
Silvicultural interventions	King et al. (1999)
Biotic disturbance	Hogg et al. (2005)
Environmental factors	
Ecoregion or biome, range extent, climate, elevation	Chave et al. (2005); Wang et al. (2006)
Site quality	Wirth et al. (2004)
Resource availability	Gower et al. (1992); Muller-Landau et al. (2006)
Data-modeling factors	
Field data collection, measurement protocols, and sampling strategies	Brown (2002)
The modeling process, i.e., statistical methods and models and issues at large scales when comparing estimates from site-level and generalized models	Jenkins et al. (2003) (see Table 8); (Zianis and Mencuccini (2004)

local and generalized regional equations across all species (p values ranged from 0.06 to 0.98). These results verify that statistically similar biomass and volume predictions were possible from the generalized regional equations when compared with the local equations for this boreal region of west-central Canada.

The magnitude of prediction bias is notably a function of original tree size and its predicted value. Local site biomass equations were the least variable throughout the DBH range (Fig. 3a). Similarly, there were little differences in $\overline{\text{Bias}}$ up to approximately 20 cm DBH, with trends of increasing bias for larger trees across all species. This pattern was similar between local site and generalized regional volume equations, with $\overline{\text{Bias}}$ for large trees >30 cm DBH, resulting in larger differences for 5 of the 10 species (black spruce, lodgepole pine, jack pine, balsam poplar, and white birch) (Fig. 3b). Prediction bias for both biomass and volume becomes a consideration when tree size is considered, but this only appears to be a factor for trees >30 cm DBH.

Discussion

Study advancement

This study has quantified the differences in predictive performance for tree biomass and volume derived from local, generalized regional, and generalized national equations that are applicable to west-central Canada. Average prediction errors computed from PRESS increased from local site to generalized regional and generalized national equations. On average, across all species and sites, biomass predictions between local site and generalized regional equations were statistically similar. For 5 of the 10 species (white spruce, lodgepole pine, jack pine, alpine fir, and white birch), there were no statistical differences in biomass predictions among the three sets of equations (Table 4). The magnitude of $\overline{\text{Bias}}$ did not vary greatly by tree size, but this was only obvious for trees <30 cm DBH in size. Over- or under-predictions were more variable for trees >30 cm DBH, and this was at-

tributed, in part, to the relatively small numbers of trees sampled in the larger diameter classes (unpublished data). These results appear consistent with recommendations that generalized equations are useful for larger scale applications, whereas, for site-specific applications, locally derived equations will normally produce more precise estimates (Jenkins et al. 2003). However, given the relative unavailability of local equations, generalized regional equations can produce relatively robust, average local predictions throughout the region investigated in this study.

The expected level of prediction error when using local versus generalized equations is not generally known nor has it been commonly reported in the literature. The relative lack of availability in, or costs associated with, establishing a coherent data set of raw biomass measurements across large geographic extents have created limitations in the degree to which prediction errors can be reliably computed. One approach for overcoming such data limitations is to create new generalized equations by generating “pseudo-data” from a variety of published equations, followed by fitting a new allometric function through these data (Jenkins et al. 2003). Whereas this approach would provide data for generalized equations where biomass data are unavailable, the lack of independent site-level data against which to compute actual prediction errors limits the ability to assess equation predictions. Other studies (Schroeder et al. 1997; Haripriya 2000) have used original biomass data to generate generalized allometric equations with reliance on regression statistics as an indicator of equation performance. From the perspective of generalized equations that may be applied over a wide range of environmental and biological conditions, regression fit statistics may provide an overly optimistic indication of prediction performance when estimating tree biomass at new sites. For example, while FI and SEE values for the generalized regional (Table 2) equations were similar to the R^2 and RMSE values reported for the national (Lambert et al. 2005) equations, there were clear and quantifiable differences in the APE and $\overline{\text{Bias}}$ error statistics com-

puted. The leave-one-out cross-validation process with the PRESS statistic resulted in estimates of APE that were used to assess differences among the local and generalized equations by species. As such, this study has demonstrated the importance of computing prediction error statistics for assessing the applicability and accuracy of biomass and volume equations.

Possible factors affecting estimates from biomass and volume equations

Increasing the spatial extent over which sample data were pooled resulted in a relative increase in prediction error. This was likely related to the increasing variety and number of factors that could influence tree growth and allometry over such a large geographic extent (Table 5). However, results indicated that the impact of data pooling on the predictive accuracy of biomass and volume equations varied by species and their relative sample size. Results from national equations differed significantly in calculated Bias among the five species (Black spruce, Tamarack, Balsam fir, Trembling aspen, and Balsam poplar) compared with local and regional estimates. For example, deciduous species were observed to be more difficult in terms of generalized modeling than conifer species. This difficulty may be attributed to the more varied crown morphologies of deciduous species when compared with coniferous species (Burns and Honkala 1990), the wide geographic distribution over which trembling aspen, balsam poplar, and white birch occur, their occupation over a wide variety of sites of differing productive potential, and the several means by which they can reproduce vegetatively (Peterson and Peterson 1992).

Errors are potentially introduced whenever an allometric equation is used, regardless of method and spatial scale (Jenkins et al. 2003). Particularly at larger spatial scales, it is difficult to fully account for all of the factors that may influence tree allometry as they are inherently integrated. Based on the literature, variations in diameter-biomass/volume allometry may be influenced by biological and environmental factors that not only occur across multiple spatial and temporal scales, but they may also be modulated by the sampling and data-modeling techniques employed (Table 5). These factors will influence some species more so than others because of a given species' ecological requirements and its interactions with other species across its native range (Muller-Landau et al. 2006). Accounting for some of these factors (Table 5) may necessitate use of different statistical and modeling approaches. One approach is to include other predictor variables, in addition to diameter, into allometric equations (Ung et al. 2001). Tree height has often been used as an additional predictor variable for allometric modeling of aboveground tree biomass (Vallet et al. 2006) because it is an indicator of site quality (eg., site index), and when combined with DBH, it becomes an indicator of tree taper. A few studies have proposed using wood density as a covariable (Chave et al. 2005), whereas others have used stand age (Bond-Lamberty et al. 2002). Other approaches have included stratifying a large sampling area by ecozone or forest type to reduce the influence of environmental variability (Chave et al. 2005). The differences in biomass predictions and related errors calculated in this study were likely due to one or several of the biological, environmental, and data-modeling factors

(Table 5) for each species, although attempting to isolate their effects would be difficult and was beyond the scope of this study. More sophisticated statistical modeling procedures, such as mixed-effects modeling (Pinheiro and Bates 2000) or hierarchical Bayesian approaches (Gelman et al. 2004), may need to be employed to tease out the various multiscale influences on biomass allometry. Ultimately, the ability to carry out such statistical analyses will be determined by the availability, quantity, and quality of available biomass sample data for a given species' range.

Conclusions

This study offers a set of generalized, regional equations for 10 species within the boreal forest region of west-central Canada that can be used to predict both total aboveground tree biomass and volume from tree diameter. Using a cross-validation process to compute prediction errors, biomass and volume estimates from the generalized regional equations were considered unbiased and not statistically different from local site estimates. Although average prediction errors, computed from PRESS, did increase with increasing equation generalization, there were no differences in prediction bias between the local and generalized regional or national equations for 5 of the 10 species analyzed (white spruce, lodgepole pine, jack pine, alpine fir, and white birch). Whereas the literature has reported numerous biological, environmental, and data-modeling factors that could affect tree allometry, particularly at increasing levels of geographic application, further work would be needed to more precisely quantify their effects.

The value of these generalized tree equations lies in their applicability to produce estimates of aboveground biomass and total volume at sample plots that would be comparable to those that could be estimated from local site equations. Tree level biomass estimates are a prerequisite for deriving stand level estimates and future work will attempt to further identify and address issues associated with scaling tree to stand level estimates. Working towards the mapping of aboveground biomass and stand volume will support national efforts for characterizing these attributes in the assessment and monitoring of C stocks for this region of Canada's forests.

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